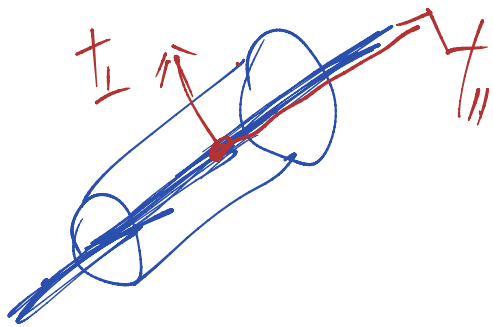


NOTEBOOK

全向轮

轮子受力分析



对轮子受力分析

仅沿轴向的分力 $F_{||}$ 能提供有效力为静摩擦。

$F_{\perp}$ 则会使轮子发生滚动为滚动摩擦。

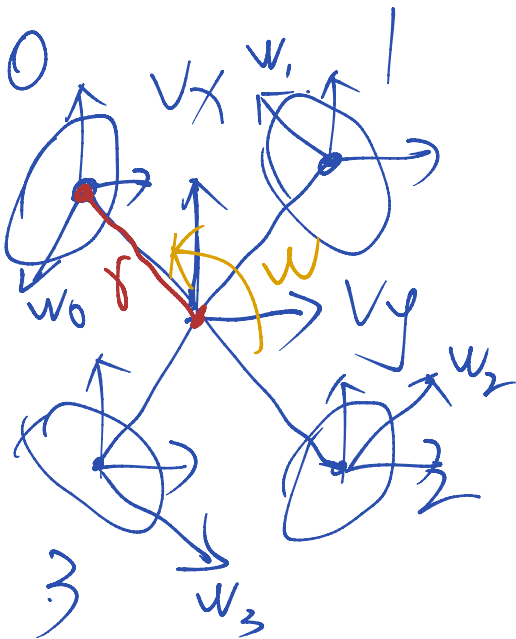
半径为 r 车体中心到轮子中心距

规定底盘正方向 $\uparrow v_x \rightarrow v_y$ 为正
 $\uparrow \omega$ 为正。

每个轮子的速度为 $\vec{v}_n$

$$\vec{v}_n = \vec{v}_{nx} + \vec{v}_{ny} + \vec{\omega}_n$$

显然: $\vec{v}_{nx} = \vec{v}_x$ $\vec{v}_{ny} = \vec{v}_y$



$$\therefore \vec{v}_0 = (v_x - \omega r \frac{r_2}{2}) \vec{i} + (v_y - \omega r \frac{r_2}{2}) \vec{j}$$

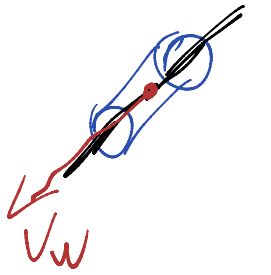
$$\text{简写: } \vec{v}_0 = (v_x - \omega r \frac{r_2}{2}, v_y - \omega r \frac{r_2}{2})$$

$$\vec{v}_1 = (v_x + \omega r \frac{r_2}{2}, v_y - \omega r \frac{r_2}{2})$$

$$v_2 = (v_x + \omega r \frac{r_2}{2}, v_y + \omega r \frac{r_2}{2})$$

$$v_3 = (v_x - \omega r \frac{r_2}{2}, v_y + \omega r \frac{r_2}{2})$$

显然主向轮 轴子的轴向与每个轮子 W 的速度
分量平行



第0个轮子

研究轴子轴向的母速度分量为多少

$$\text{不妨设 } \vec{e}_{||n} = \frac{\vec{V}W_n}{|\vec{V}W_n|}$$

$$\begin{aligned} V_{||0} &= (V_x - WR\frac{\sqrt{2}}{2})\cos 135^\circ + (V_y - WR\frac{\sqrt{2}}{2})\cos 135^\circ \\ &= -\frac{\sqrt{2}}{2}V_x - \frac{\sqrt{2}}{2}V_y + WR \end{aligned}$$

$$\begin{aligned} V_{||1} &= (V_x + WR\frac{\sqrt{2}}{2})\cos 45^\circ + (V_y - WR\frac{\sqrt{2}}{2})\cos 135^\circ \\ &= \frac{\sqrt{2}}{2}V_x - \frac{\sqrt{2}}{2}V_y + WR \end{aligned}$$

$$\begin{aligned} V_{||2} &= (V_x + WR\frac{\sqrt{2}}{2})\cos 45^\circ + (V_y + WR\frac{\sqrt{2}}{2})\cos 45^\circ \\ &= \frac{\sqrt{2}}{2}V_x + \frac{\sqrt{2}}{2}V_y + WR \end{aligned}$$

$$\begin{aligned} V_{||3} &= (V_x - WR\frac{\sqrt{2}}{2})\cos 135^\circ + (V_y + WR\frac{\sqrt{2}}{2})\cos 45^\circ \\ &= -\frac{\sqrt{2}}{2}V_x + \frac{\sqrt{2}}{2}V_y + WR \end{aligned}$$

那 $V_{||}$ 与 $V_{\text{轴}W}$ (电机转速) 之间的关系

$Vw \uparrow$ 车子摩擦力 f .

$Vw = \alpha F$ 替认为是 α 的关系.

$$F \Delta t = m \Delta V_{||}$$

$$F = \frac{m}{\Delta t} \Delta V_{||}$$

$$\therefore Vw = \alpha \frac{m}{\Delta t} \Delta V_{||}$$

不妨设 $\alpha \frac{m}{\Delta t} = k$ $v_y + v_x$

则 $Vw = k \Delta V_{||}$

则 $Vw_0 = k \left(-\frac{r_2}{2} v_x - \frac{r_2}{2} v_y + w_x \right)$

$$Vw_1 = k \left(\frac{r_2}{2} v_x - \frac{r_2}{2} v_y + w_x \right)$$

$$Vw_2 = k \left(\frac{r_2}{2} v_x + \frac{r_2}{2} v_y + w_x \right)$$

$$Vw_3 = k \left(-\frac{r_2}{2} v_x + \frac{r_2}{2} v_y + w_x \right)$$

$$\begin{array}{c}
 \begin{array}{cc}
 V_y & V_x \\
 \left[\begin{array}{cc}
 -\frac{\kappa_2}{2} & -\frac{\kappa_2}{2} \\
 -\frac{\kappa_2}{2} & \frac{\kappa_2}{2} \\
 \frac{\kappa_2}{2} & \frac{\kappa_2}{2} \\
 \frac{\kappa_2}{2} & -\frac{\kappa_2}{2}
 \end{array} \right]
 \end{array}
 &
 \begin{array}{c}
 w_r \\
 \left[\begin{array}{c} 1 \\ 1 \\ 1 \\ 1 \end{array} \right]
 \end{array}
 \end{array}
 \xrightarrow{\times -1}
 \begin{array}{c}
 \begin{array}{cc}
 \left[\begin{array}{cc}
 \frac{\kappa_2}{2} & \frac{\kappa_2}{2} \\
 \frac{\kappa_2}{2} & -\frac{\kappa_2}{2} \\
 -\frac{\kappa_2}{2} & \frac{\kappa_2}{2} \\
 -\frac{\kappa_2}{2} & \frac{\kappa_2}{2}
 \end{array} \right]
 \end{array}
 &
 \begin{array}{c}
 \left[\begin{array}{c} -1 \\ -1 \\ -1 \\ -1 \end{array} \right]
 \end{array}
 \end{array}
 \end{array}$$

这里全乘-1是因为
电机正电流时 V_w 与我们假设相反

∴ 最终

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} \frac{r_2}{2} & \frac{r_2}{2} & -1 \\ \frac{r_2}{2} & -\frac{r_2}{2} & -1 \\ -\frac{r_2}{2} & \frac{r_2}{2} & -1 \\ \frac{r_2}{2} & \frac{r_2}{2} & -1 \end{bmatrix} \begin{bmatrix} V_x \\ V_y \\ V_w \end{bmatrix}$$

那公式前面的k呢?

$V_w \uparrow$ 轮子摩擦力↑

$V_w = \alpha F$ 暂认为是正比的关系

$$F \Delta t = m \Delta V_{||}$$

$$F = \frac{m}{\Delta t} \Delta V_{||}$$

$$\therefore V_w = \alpha \frac{m}{\Delta t} \Delta V_{||}$$

$$\text{不妨设 } \alpha \frac{m}{\Delta t} = k \quad v_y + v_x$$

$$\text{则 } V_w = k \Delta V_{||}$$

在你 PDF 第4页的推导中:

$$\begin{aligned} V_w &= \Delta \frac{m}{\Delta t} \Delta V_i \\ \text{不设 } \Delta \frac{m}{\Delta t} &= k \\ V_w &= k \Delta V_i \end{aligned}$$

这个k在物理推导和实际代码工程中,有着完全不同的含义。下面为你彻底梳理:

1. 物理推导层面的k

在你的推导里, $k = \frac{m}{\Delta t}$ (这里的 Δ 作为比例符号)。这是基于动量定理 $F \Delta t = m \Delta V_i$ 推导出来的。它代表了全向轮与地面接触时的等效动力学增益。这个值受摩擦系数、地面材质、轮子材质、电机扭矩等影响,在理论推导中它是个常数,但在实际工程中,你没法通过计算得出一个精确的数值。

```
• wheel_speed[0] = (SQRT2 * vy_set + SQRT2 * vx_set - MOTOR_DISTANCE_TO_CENTER * wz_set);  
• wheel_speed[1] = (SQRT2 * vy_set - SQRT2 * vx_set - MOTOR_DISTANCE_TO_CENTER * wz_set);  
• wheel_speed[2] = (-SQRT2 * vy_set - SQRT2 * vx_set - MOTOR_DISTANCE_TO_CENTER * wz_set);  
• wheel_speed[3] = (-SQRT2 * vy_set + SQRT2 * vx_set - MOTOR_DISTANCE_TO_CENTER * wz_set);
```

